Q2: Adversarial Search

- Due 12 Jan at 12:30
- Points 5
- Questions 5
- Available after 4 Nov 2025 at 17:00
- Time limit 60 Minutes
- Allowed attempts 5

This quiz is no longer available as the course has been concluded.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	2 minutes	5 out of 5

ⓘ Correct answers are hidden.

Score for this attempt: 5 out of 5
Submitted 30 Nov 2025 at 19:06
This attempt took 2 minutes.

Question 1
1 / 1 pts
What does alpha-beta pruning do?

- (•) Ignores branches that will not be chosen by an optimal player
- ( ) Ensures the max player will always win
- ( ) Lets the algorithm search 10 times deeper than minimax alone
- ( ) Checks all possible nodes in the tree

Question 2
1 / 1 pts
Which of the following describes chess?

- ( ) Imperfect information and deterministic
- ( ) Imperfect information and chance
- (•) Deterministic and perfect information

- Perfect information and chance

Question 3

1 / 1 pts

In a zero-sum game, a terminal state gives MAX a utility of **+5**. Which of the following is **true**?

- MIN's utility must be 0
- MIN's utility could be any value less than 5
- This tells us nothing about MIN's utility
- MIN's utility must be −5

Question 4

1 / 1 pts

Suppose we are at a MIN node in a game tree with child node values of `[5,2,7,4]`. Which value will MIN return to its parent node?

- 5 — because it is the first value evaluated and no better options are guaranteed
- 4 - because it's the lowest integer not exceeding the average utility of the children
- 2 — because MIN selects the lowest utility among its children to minimize MAX's gain
- 7 — because a rational opponent would never allow the MIN player to access higher values later
- 7 — because MIN selects the highest utility for the opponent
- 2 — because the value farthest from the median is often safest when minimizing risk
- 4.5 - because it is the median of the set
- 4 — because it is closest to the average and thus represents a balanced choice

Question 5

1 / 1 pts

Suppose you are using minimax with alpha-beta pruning in a deterministic, zero-sum game. MAX is the root and expands its children in an arbitrary (non-optimal) order. What is the most likely effect on the search process?

- The algorithm still guarantees optimal decisions, but may explore more nodes due to suboptimal pruning.
- MAX will fail to find the optimal move because alpha will never exceed beta.
- Pruning will improve because MIN will be forced to cut branches earlier
- Pruning efficiency will remain the same

Quiz score: 5 out of 5